SELECTED WORK

Iyinoluwa Don-Taiwo

Machine learning engineer
and data scientist

Projects with their data, results and limits.



PROJECT DECK / 2026

MACHINE LEARNING DATA SCIENCE

Classifying an ACL tear from a full MRI volume

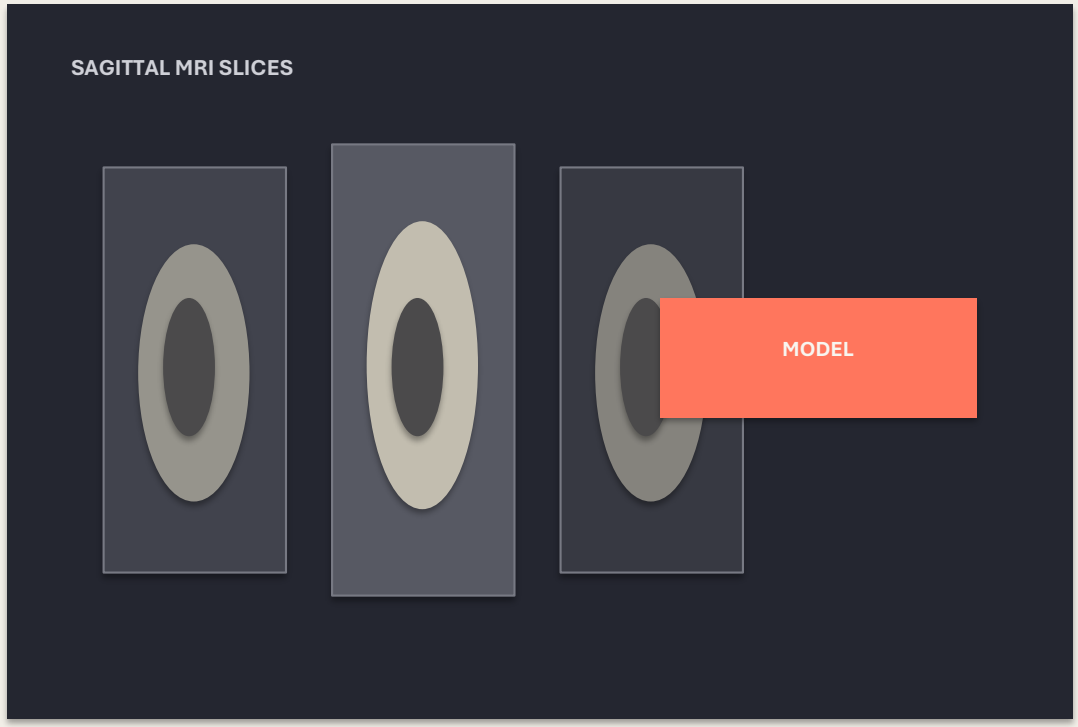
A ResNet18 transfer-learning model reads sagittal knee MRI slices. The Streamlit demo accepts NumPy volumes, with optional DICOM and NIfTI readers.

0.9719

reported AUC-ROC

120-exam validation set / Stanford MRNet: 1,370 exams

Repository: github.com/IyinoluwaDon/acl-tear-detection



Schematic only. Research prototype, not a diagnostic tool. The reported result has not been independently reproduced here.

Does the product photo match its description?

I used downloaded Rakuten France listings to compare text-only, image-only and fused models. The task pairs real listings with deliberately swapped images to create mismatch examples.

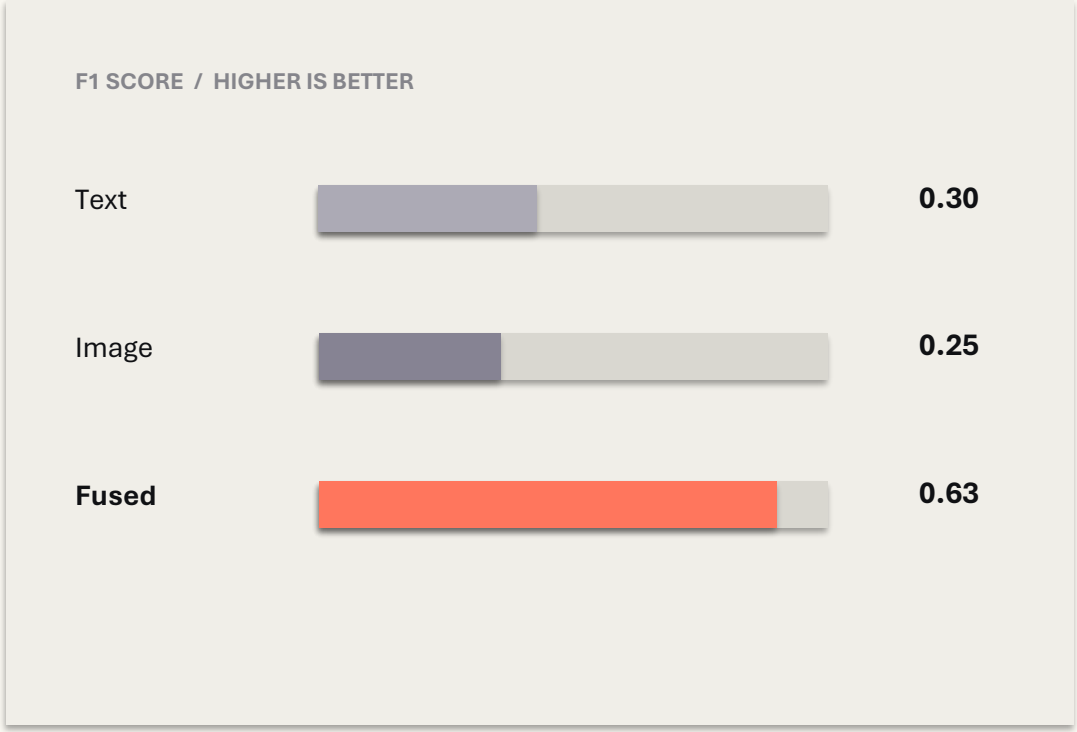
20,000

product listings in the subset

25%

images swapped for proxy labels

Repository: github.com/IyinoluwaDon/ecommerce-listing-consistency-detector



The mismatch label is a proxy signal, not a general moderation or policy decision.

Seasonal inputs lowered forecast error on the holdout

The research notebook uses Sentinel-2 NDVI features for Omo Forest Reserve and compares an LSTM with seasonal features against an NDVI-only baseline.



MODEL NOTEBOOK

The hosted dashboard uses illustrative values. The notebook contains the model and reported evaluation.

Repository: github.com/IyinoluwaDon/Omo-Forest-Deforestation-Forecasting

Small holdout with interpolated months. NDVI change alone does not establish deforestation.

Fine-tuning Gemma 2 for an African folktale challenge

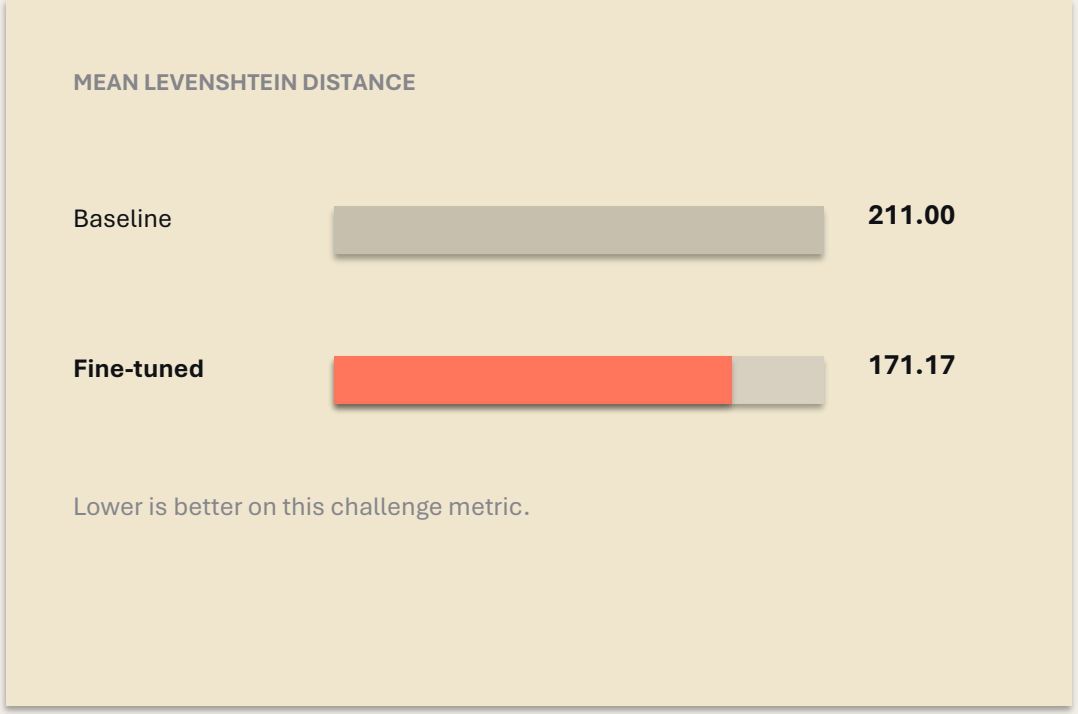
I led Team Ethopie, a nine-person group, through the TRI AI Saturdays Cohort 10 capstone. I co-developed the Gemma 2 2B LoRA notebook with one teammate.

3rd

place in the cohort capstone

24 synthetic folktales / 38 prompt-story pairs

Repository: github.com/IyinoluwaDon/C10-team-Ethopie



Synthetic benchmark. The score does not assess story quality or cultural fidelity.

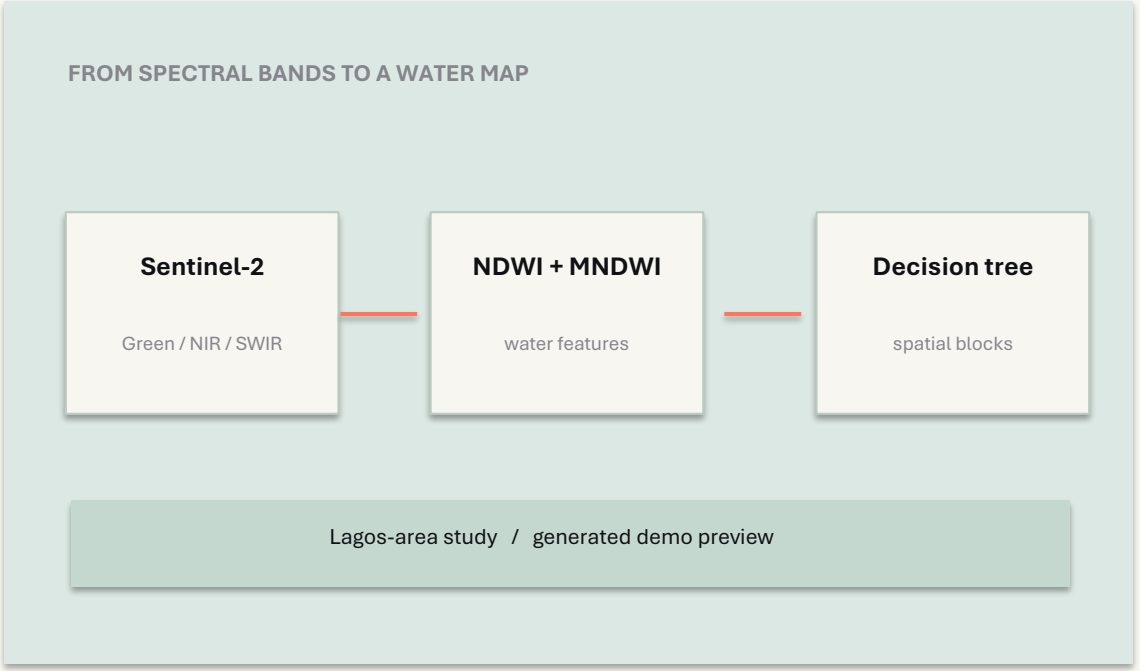
Mapping surface water with Sentinel-2 indices

The Python project derives two water indices from Sentinel-2 bands, then fits a decision tree against historical surface-water labels. Spatial blocks separate evaluation areas.

Included

Threshold baselines
Documented map classes
Optional Earth Engine export

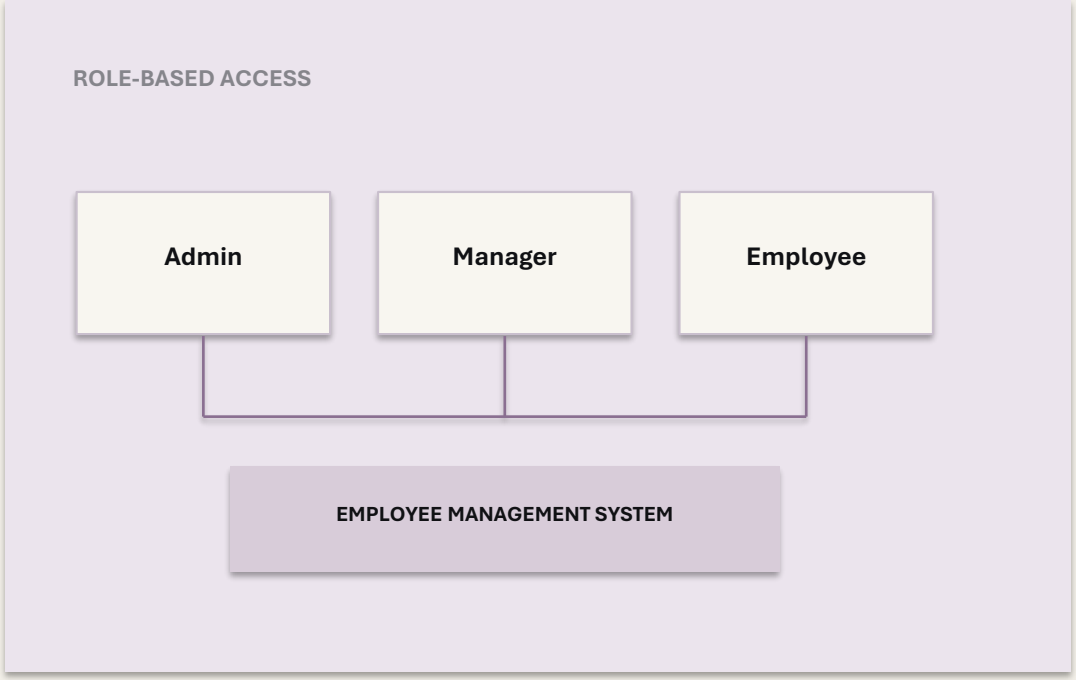
Repository: github.com/IyinoluwaDon/Water-body-detection--Sentinel-2



Generated demo data checks the pipeline only. Accuracy on real imagery still needs independent reference checks.

Employee records, attendance and payroll in one PHP app

Shebamiles EMS covers employee records, departments, attendance, leave, payroll and performance reviews. Each account has an admin, manager or employee role.



Security work

Prepared SQL statements
Password hashing and CSRF checks
First-admin setup through a CLI command

Repository: github.com/IyinoluwaDon/Shebamiles-EMS

PHP checks and auth tests pass. Live MySQL integration has not been run in this workspace.

Checking churn comparisons in an Excel workbook

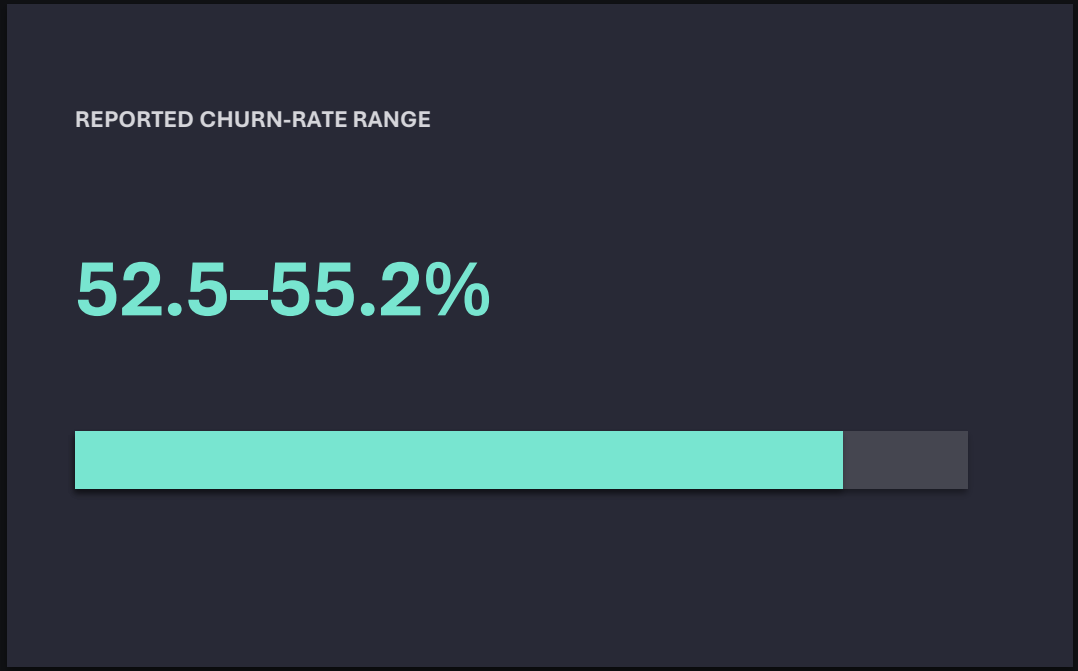
The workbook summarizes 500 customer records with PivotTables and statistical tests. I preserved the input sheet and added a verifier that checks row counts, identifiers, missing values and the reported calculations.

F = 0.9898

Credit-score ANOVA / p = 0.3724

Salary t-test / p = 0.42

Repository: github.com/IyinoluwaDon/bank_churn_analysis



Sample source and license are undocumented. Descriptive results do not establish causes or general bank-wide churn rates.

Twelve read-only queries over an HR database

A small SQLite analysis validates database integrity, calculates tenure relative to review dates, runs twelve named queries and exports the results as CSV and JSON.

1,000

employee records

238

Engineering employees

Repository: github.com/IyinoluwaDon/HR-Data-Analysis-with-SQL

READ-ONLY QUERY OUTPUT

INTEGRITY	PASS
FOREIGN KEYS	PASS
NAMED QUERIES	12
EXPORTS	CSV + JSON

Database source and field definitions are undocumented. Findings describe only the supplied sample.

THANK YOU

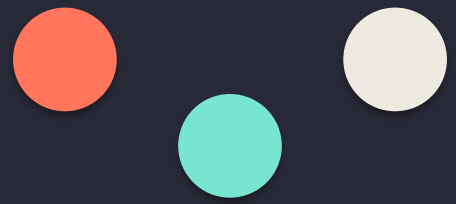
Get in touch

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Code, demos and project notes are linked from the portfolio.

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